



Antidepressant magic mushrooms

Research has shown that psilocybin still works as an antidepressant without the "high".
<http://digit.in/may21-27>



Whitest-ever paint

"Whitest-ever" paint can reflect 98% of sunlight, radiating infrared heat into space.
<http://digit.in/may21-28>

Image of 2I/Borisov captured by ESO's Very Large Telescope when the comet was close to the Sun in December 2019.

Credit: ESO/O Hainaut

Alien Stardust

Observations of a strange little comet discovered by an amateur astronomer is ushering in a new era of astronomy

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The object designated as 2I/Borisov, is a comet, one of the many small objects discovered by Gennadiy Borisov, an amateur astronomer. Comet hunting is a popular sport for citizen scientists, but what makes Borisov so special is the "I" in the designation which stands for "interstellar". It is the second known visitor to the Solar System after 'Oumuamua, and we covered the discovery of the comet in Space Age, in the November 2019 issue of Digit. Borisov was discov-

ered earlier in its visit, unlike 'Oumuamua which was on its way out when it was spotted, allowing for a much longer observation period using all the premiere observatories in the world, which would turn towards the same object discovered by a 65 cm long home built telescope, with observations expected to continue till 2020. Astronomical instruments were in position, and had plenty of time to characterise and study this object, and the results from those investigations have been coming in, increasing our understanding of the object and throwing up new mysteries.

Hubble observed the object at its closest approach to the Sun, in December 2019. This observation allowed astronomers to fix the upper limit of the size of the nucleus of the comet, which is a very important measurement that affects all the other studies on the object. David Jewitt, a UCLA professor of planetary science and astronomy whose team captured the image said, "Hubble gives us the best upper limit of the size of

comet Borisov's nucleus, which is the really important part of the comet. Surprisingly, our Hubble images show that its nucleus is more than 15 times smaller than earlier investigations suggested it might be. Our Hubble images show that the radius is smaller than half a kilometer. Knowing the size is potentially useful for beginning to estimate how common such objects may be in the solar system and our galaxy. Borisov is the first known interstellar comet, and we would like to learn how many others there are."

At the perihelion, when the comet was closest to the Sun, another observatory was observing Borisov, ALMA, or the Atacama Large Millimeter/submillimeter Array. These observations allowed astronomers to study the material outgassing from the comet, something that 'Oumuamua did not do. As the comets spend most of their time relatively far away from their stars, the material within them preserve the clue to the conditions during the formation of the stellar